

Mathematical Analysis

- Completeness of $\mathbb{R}$
- Uniform continuity
- Uniform convergence

Chapter 1: Real Number ($\mathbb{R}$)

§ 1.1: Concepts:

- $\mathbb{R} = (-\infty, \infty)$
 $\emptyset = \text{empty set}$
 A is a set, a is an element: $a \in A$ or $a \notin A$
- $A \subseteq B$: A is a subset of B .
 def: for any $a \in A$, we have $a \in B$
- $\forall$: for any
 $\exists$: there exists
- $A \subset B$ if $A \subseteq B$ and " A is a proper subset of B "
 $\Rightarrow$ " A is a proper subset of B " $\Leftrightarrow$ there is $b \in B$ where $b \notin A$
- $A = B$: $A \subseteq B$ and $B \subseteq A$

Set operation

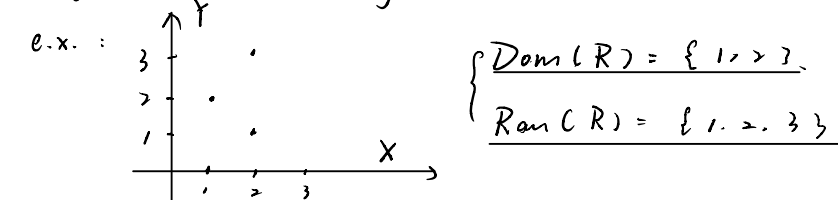
- A, B : "is" definition
- $A \cup B := \{x: x \in A \text{ or } x \in B\}$
 - $A \cap B := \{x: x \in A \text{ and } x \in B\}$
 - $A \setminus B := \{x: x \in A \text{ and } x \notin B\}$

Cartesian Product:

- X, Y are sets. $X \times Y := \{(x, y): x \in X, y \in Y\}$
 $(x, y), (z, w) \in X \times Y, (x, y) = (z, w) \Leftrightarrow x = z \text{ and } y = w$
 $x \times y = \emptyset$

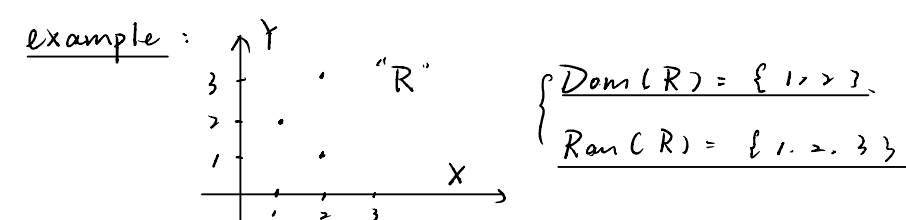
Function: Let X and Y be sets. Any subset $R \subseteq X \times Y$ is called relation on $X \times Y$

domain: $\text{Dom}(R) := \{x \in X: (x, y) \in R \text{ for some } y \in Y\} \subseteq X$
 range: $\text{Ran}(R) := \{y \in Y: (x, y) \in R \text{ for some } x \in X\} \subseteq Y$



★ Def: a function f from X into Y is a relation on $X \times Y$

such that $\text{Dom}(R) = X$ and $\forall x \in X$, there is a unique $y \in Y$ with $(x, y) \in f$



Is relation R a function?

- No! since: 1. $\text{Dom}(R) \neq X$.
 2. $\exists x \in X$, there exists more than one $y \in Y$, $(x, y) \in R$.

Notation

Suppose f is a function from X into Y

We write $f: X \rightarrow Y$

if $(x, y) \in f$, we call y the value of f at x , notation: $y = f(x)$

or $f: x \mapsto y$, we call x is a preimage of y

if $f: X \rightarrow Y$, we say f is defined on X and Y is the codomain of f

$\text{Ran}(f) := \{y \in Y: y = f(x) \text{ for some } x \in X\}$

$= \{f(x): x \in X\} \subseteq Y$

$y \in \text{Ran}(f) \Leftrightarrow \exists x \in X, y = f(x)$

Inverse function:

if $\text{Ran}(f) = Y$ and for any $y \in Y$ there is a unique $x \in X$ such that $f(x) = y$

then we say $f: X \rightarrow Y$ has an inverse

and define $f^{-1}: Y \rightarrow X, y \mapsto x$

* two function f, g are the same if $\text{Dom}(f) = \text{Dom}(g)$ and $\text{Ran}(f) = \text{Ran}(g)$ and $\forall x, f(x) = g(x)$

Real functions: $f: \mathbb{R} \rightarrow \mathbb{R}$

- $e^x: \mathbb{R} \rightarrow (0, \infty)$
 $\log x: (0, \infty) \rightarrow \mathbb{R}$ inverse.
- trig function: $\sin x, \cos x, \tan x, \cot x, \dots$
- power function: $x^a, x > 0, a \in \mathbb{R}$

1.2 Ordered-field axioms

What is $\mathbb{R}$?

define: $\mathbb{N} = \{1, 2, 3, \dots\}$ Natural Numbers

$\mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$ Integers

$\mathbb{Q} = \{\frac{m}{n}: m, n \in \mathbb{Z}, n \neq 0\}$ rational numbers.

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ " $\mathbb{R} \notin \mathbb{Q}$ "

$\mathbb{Q} = \mathbb{R}/\mathbb{Q}$ "Set of irrational numbers"

What is $\mathbb{R}$?

① $\mathbb{R}$ is a field satisfying field axiom:

Field axiom: $+$, $-$, $\times$: three operations satisfy natural properties

$a+b = b+a, (a+b)+c = a+(b+c), a \cdot 0 = a \cdot 1 = 1$

② There is an order " $<$ " on $\mathbb{R}$ that is the order axiom:

$\forall a, b \in \mathbb{R}, a < b$ or $a > b$ or $a = b$

natural properties: $a < b \Rightarrow a+c < b+c \quad \forall c \in \mathbb{R}$
 $a < b \Rightarrow ac < bc \quad \forall c \in \mathbb{R}, c > 0$
 $a < b \Rightarrow ac < bc \quad \forall c \in \mathbb{R}, c < 0$

$a > b, b > c \Rightarrow a > c$

$a = b \Leftrightarrow a < b$ or $a = b$

remark: $\mathbb{N}, \mathbb{Z}$ satisfies:

- $m, n \in \mathbb{Z} \Rightarrow m+n, mn \in \mathbb{Z}$
- $n \in \mathbb{Z}$ then $n \in \mathbb{N} \Leftrightarrow n \geq 1$
- there is no $n \in \mathbb{Z}$ such that $0 < n < 1$ (i.e. $\mathbb{Z} \cap (0, 1) = \emptyset$)

Absolute Value

$a \in \mathbb{R}, |a| := \begin{cases} a, & a \geq 0 \\ -a, & a < 0 \end{cases}$

Property: (1) $|a \cdot b| = |a| \cdot |b|$
 (2) $|a| \leq M \Leftrightarrow -M \leq a \leq M$
 (3) $|a| = |-a| \Rightarrow |a-b| = |b-a|$
 (4) $|a+b| \leq |a|+|b|$ "triangle inequality"

① $|a|-|b| \leq |a-b|$

proof: $|a|, |b| \leq |a-b|+|b|$

thus, $|a|-|b| \leq |a-b|$

interchanging a, b : $|b|-|a| \leq |b-a| = |a-b|$

② $|a|-|b| \leq |a-b|$

By ①, ②: $-|a-b| \leq |a|-|b| \leq |a-b|$

therefore $||a|-|b|| \leq |a-b|$

Interval on $\mathbb{R}$

$a, b \in \mathbb{R}$

- $[a, b] := \{x \in \mathbb{R}: a \leq x \leq b\}$
- $(a, b) := \{x \in \mathbb{R}: a < x < b\}$
- $[a, b) := \{x \in \mathbb{R}: x \geq a, x < b\}$
- $(a, b] := \{x \in \mathbb{R}: x > a, x \leq b\}$

§ 1.3 Completeness Axiom

Def: $\emptyset \neq E \subseteq \mathbb{R}$ (E is a nonempty subset of $\mathbb{R}$)

i) E is called bounded above $\Leftrightarrow \exists M \in \mathbb{R}$ such that $\forall x \in E, x \leq M$

e.g. $E = (1, 5)$ then $M = 7$ is an upper bound, E is bounded above.

Remark: ① if M is an UB, so is $M+1 \Rightarrow$ Upper Bound is not unique if exists.

② UB may not exist (e.g. $E = \mathbb{R}$)

ii) A number $g \in \mathbb{R}$ is called a supremum of E

if and only if: ① g is an upper bound of E

② $g \leq M$ for any upper bound M of E (the smallest UB)

if E has a supremum s . We write $g = \sup E$

e.x. $E = [0, 1]$ prove that $\sup E = 1$

proof: $\forall x \in [0, 1], x \leq 1$ so 1 is an UB.

Let M be an UB of E , then $M \geq x \quad \forall x \in [0, 1] \Rightarrow M \geq 1$

$\therefore \sup E = 1$ (prove by definition)

Remark: ① if E has a supremum, then it has only one supremum (uniqueness)

proof: if E has two supremum s, t .

then by def of supremum: $s \leq t$ and $t \leq s$, therefore: $s = t$

Theorem: if E has a supremum, let $g = \sup E$

Then $\forall \epsilon > 0, \exists a \in E$ such that $a \geq g - \epsilon$

Proof: suppose not, then:

$\exists \epsilon > 0, \forall a \in E, a \leq g - \epsilon$

then $g - \epsilon$ is an UB of E .

Since $g = \sup E$ and $g - \epsilon < g$, contradiction!

Theorem: if $E \subseteq \mathbb{Z}$ has a supremum then $\sup E \in \mathbb{Z}$

proof: $\frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots$

"if $\sup E$ is not an integer we can always shift $\sup E$ left"

Completeness Axiom: $\emptyset \neq E \subseteq \mathbb{R}$

"if E is bounded above, then E has a supremum i.e. $\sup E \in \mathbb{R}$ "

Archimedean Principle $\forall a, b \in \mathbb{R}, a > 0$, then $\exists n \in \mathbb{N}, b < na$

(i.e. na can be arbitrarily large as $n \in \mathbb{N}$)

Proof: if $b < a$, see n: 1

if $b \geq a$, consider $E = \{k \in \mathbb{N}: k \leq b\}$: ① $1 \in E \neq \emptyset$ since $a \leq b$

② $\forall k \in E \Rightarrow k \leq b \Rightarrow k \leq \frac{b}{a} \in \mathbb{R}$

$\Rightarrow \frac{b}{a}$ is an upper bound of E

By completeness theorem, $\exists s \in \mathbb{R}, s = \sup E$

see $n = s+1 \notin E$ then $na > b$, proved.

"Completeness axiom $\Rightarrow$ density of rationals"

e.x. $B = \{1, \frac{1}{n}, n \in \mathbb{N}\}$ prove that $\sup B = 1$

proof: (1) since $\forall a \in B, a \leq 1$, 1 is an upper bound of B .

(2) let M be UB of B , we need to show that $M \geq 1$.

if $M < 1$: By archimedean principle, $\exists n \in \mathbb{N}, n > \frac{1}{1-M}$ ($a = 1, b = \frac{1}{1-M} \Rightarrow na > b \Rightarrow n > \frac{1}{1-M}$)

$n > \frac{1}{1-M} \Rightarrow 1 - \frac{1}{n} > M$, M is not upper bound of B

Lower Bound and infima

① The set E is bounded below if there exists $m \in \mathbb{R}$ such that: $m \leq a$ for every $a \in E$

such m is a lower bound of E .

② A number $t \in \mathbb{R}$ is an infimum of E if:

1) t is a lower bound of E

2) t is the largest lower bound: $m < t$ for every lower bound m of E

written $t = \inf E$ if it exists

Supremum, infimum, negation

① def: $-E := \{-x: x \in E\}$

② E has a supremum $\Leftrightarrow -E$ has a infimum

③ E has a infimum $\Leftrightarrow -E$ has a supremum

Proof: if E has a supremum say $s = \sup E$

We need to show that $\inf(-E) = -s$

1. $\forall x \in -E, -x \leq s \Rightarrow x \geq -s, -s$ is a lower bound

2. let m be a lower bound of $-E \Rightarrow -m$ is a upper bound of E .

$-m \geq \sup E = s \Rightarrow m \leq -s$

Hence $-s$ is a infimum of $-E$

Theorem 1.2: monotonicity under inclusion

Let $A \subseteq B$ be nonempty subset of $\mathbb{R}$.

(1) if B has a supremum then A has a supremum and $\sup A \leq \sup B$

(2) if B has a infimum then A has a infimum and $\inf A \geq \inf B$

proof: (1) Every upper bound of B is a upper bound of $A \Rightarrow \sup B$ is an UB of $A \Rightarrow \sup A \leq \sup B$

(ii) $A \subseteq B \Rightarrow -A \subseteq -B \Rightarrow \sup(-A) \leq \sup(-B) \Rightarrow \inf(A) \geq \inf(B)$

The extended real number

Define: $\bar{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$

For $\emptyset \neq E \subseteq \mathbb{R}$: $\sup E = \infty$ if E is unbounded above

$\inf E = -\infty$ if E is unbounded below

For $E = \emptyset$: $\sup E = -\infty, \inf E = \infty$

$\Rightarrow$ For $A \subseteq B \subseteq \bar{\mathbb{R}}, \sup A \leq \sup B, \inf A \geq \inf B$

Mathematical induction

Every subset $E \subseteq \mathbb{N}$ has a least element: $\inf E \in E$

proof: Since $E \subseteq \mathbb{N}, -E := \{-n: n \in E\}$ is a nonempty subset of $\mathbb{Z}$

By completeness axiom: Since $-E \subseteq \mathbb{Z}, -E$ has a supremum $c \in \mathbb{R}$

By Theorem proved previously (if $E \subseteq \mathbb{Z}$ has a supremum, then $\sup E \in \mathbb{Z}$)

$\therefore \sup(-E) \in \mathbb{Z} \Rightarrow -\sup(-E) \in \mathbb{Z} \Rightarrow \inf E \in E$

Theorem: principle of mathematical induction

Let $A(n)$ be a proposition ($n \in \mathbb{N}$)

if: ① $A(1)$ is true and:

② for $n \in \mathbb{N}$ if $A(n)$ is true then $A(n+1)$ is true

proof: suppose the conclusion is false:

$E := \{n \in \mathbb{N}: A(n) \text{ is false}\} \neq \emptyset$

$m := \min E$ exists by well ordering theorem

$A(1)$ is true $\Rightarrow m \neq 1 \Rightarrow m-1 \in \mathbb{N}$

Since $m = \min E, m-1 \notin E$

$A(m-1)$ is true $\Rightarrow A(m)$ is true, which is a contradiction

Summation notation

$\sum_{j=1}^n x_j = x_1 + x_2 + \dots + x_n$

Binomial Coefficients

$n!$ is $n(n-1)\dots 1$

Binomial coefficients

$\binom{n}{k} := \frac{n!}{(n-k)!k!}$ in particular $\binom{n}{0} = 1, \binom{n}{n} = 1$

Pascal's identity

$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

$\frac{n!}{(n-k)(n-k-1)!} + \frac{n!}{(n-k-1)k!} = \frac{n!}{(n-k-1)k!} + \frac{n!}{(n-k-1)k!} = \frac{n!}{(n-k-1)k!} = \frac{(n+1)!}{(n-k-1)k!} = \binom{n+1}{k}$

Binomial Theorem:

$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$

Inverse functions

(1) one-to-one: $\forall x, y \in X, x \neq y \Rightarrow f(x) \neq f(y)$

(2) onto: $\forall y \in Y, \exists x \in X, f(x) = y$

(3) bijective: one-to-one & onto

The following are equivalent:

① $f: X \rightarrow Y$ has an inverse function $f^{-1}: Y \rightarrow X$

② f is a bijection

③ There exists a function $g: Y \rightarrow X$ such that: $g \circ f = \text{id}_X, f \circ g = \text{id}_Y$
 $"x \mapsto g(f(x)), y \mapsto f(g(y))"$

Image and Preimage

image of $E \subseteq X$: $f(E) := \{f(x): x \in E\}$

preimage of $V \subseteq Y$: $f^{-1}(V) := \{x: f(x) \in V\}$

Indexed families of sets

$\{E_i: i \in I\}, E_i \subseteq X,$